Bermuda-triangle-threat-incident-visualization

A Power BI Analytics Project

Technology: Microsoft Power BI | DAX | Excel | CSV

Project Documentation

1. Project Overview

The project focuses on developing an interactive Threat Classification & Predictive Intelligence Dashboard using Microsoft Power BI. The solution was developed through four milestones, gradually extending the analysis from basic incident reporting to geospatial intelligence, temporal and response analytics, security risk assessment, predictive intelligence, and executive decision support.

The main purpose of the project is to transform incident data into meaningful information that can help stakeholders understand threats, identify high-risk areas, evaluate response performance, and support better operational decisions.

2. Project Objectives

The major objectives of the project are:

- Analyze incident patterns and severity levels.
- Identify geographical distribution and high-risk locations.
- Study incident trends over time.
- Evaluate emergency response performance.
- Assess security risks and operational vulnerabilities.
- Predict emerging risks and future incident trends.
- Provide interactive dashboards for decision-making.

3. Datasets Used

The project uses three related datasets, connected using incident_id, allowing incident, location, and response information to be analyzed together.

Incident Dataset

- Incident ID
- Date
- Incident Type
- Category
- Casualties
- Outcome
- Latitude and Longitude

Location Dataset

- Incident ID
- City
- Latitude and Longitude

Response Dataset

- Incident ID
- Category

- Response Time
- Response Team
- Escalation
- Status

4. Tools and Technologies

- Microsoft Power BI
- DAX
- Microsoft Excel
- CSV Dataset

5. Overall Project Flow

Data Collection → Data Cleaning → Data Modeling → Visualization → Threat Analysis → Temporal & Response Analysis → Risk Assessment → Predictive Intelligence → Executive Decision Support

Milestone 1: Threat Classification & Severity Analytics

1. Objective

The first milestone established the foundation of the Power BI solution by integrating incident, location, and response data into an interactive analytical dashboard.

The dashboard was designed to understand incident patterns, severity levels, response performance, and location-based trends.

2. Data Preparation

- Verified column names and data types.
- Checked duplicate incident IDs.
- Removed unnecessary columns where required.
- Verified relationships between datasets.
- Created relationships using incident_id.
- Created a Severity column using DAX.
- Developed calculated measures for analysis.

3. Data Model

The three datasets were connected using a one-to-many relationship:

- Incident (1) → Location (*)
- Incident (1) → Response (*)

The relationships allow filters and calculations to work across the connected datasets.

4. Key Dashboard Components

The dashboard included six main KPI indicators:

Total Incidents	Critical Incidents
Threat Severity Index	Average Response Time
Total Locations	Total Escalated Incidents

5. Interactive Features

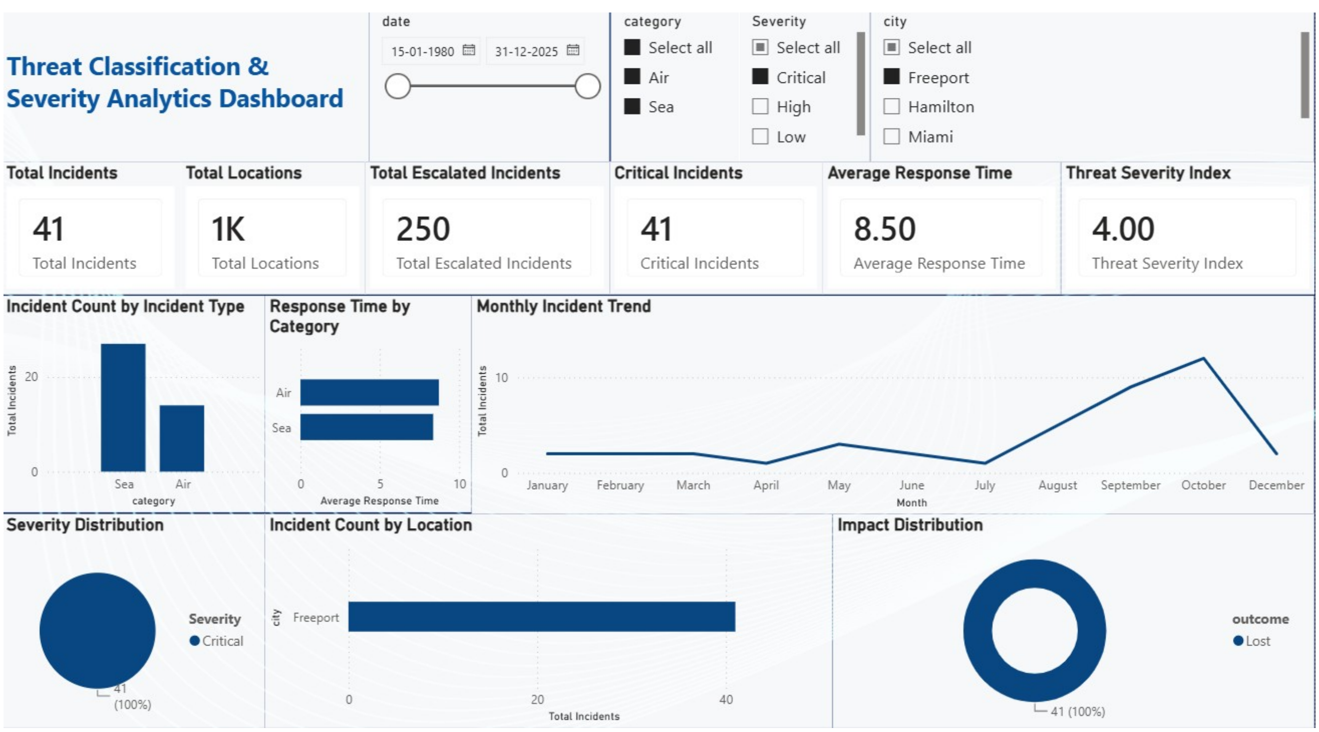
Four slicers were included:

- Date
- Category
- City

These allow users to dynamically filter the dashboard and explore incident information from different perspectives.

6. Outcome

Milestone 1 established the core Power BI data model, DAX calculations, KPI cards, charts, and interactive filtering required for the later milestones.



Milestone 2: Threat Classification & Geospatial Intelligence

Objective

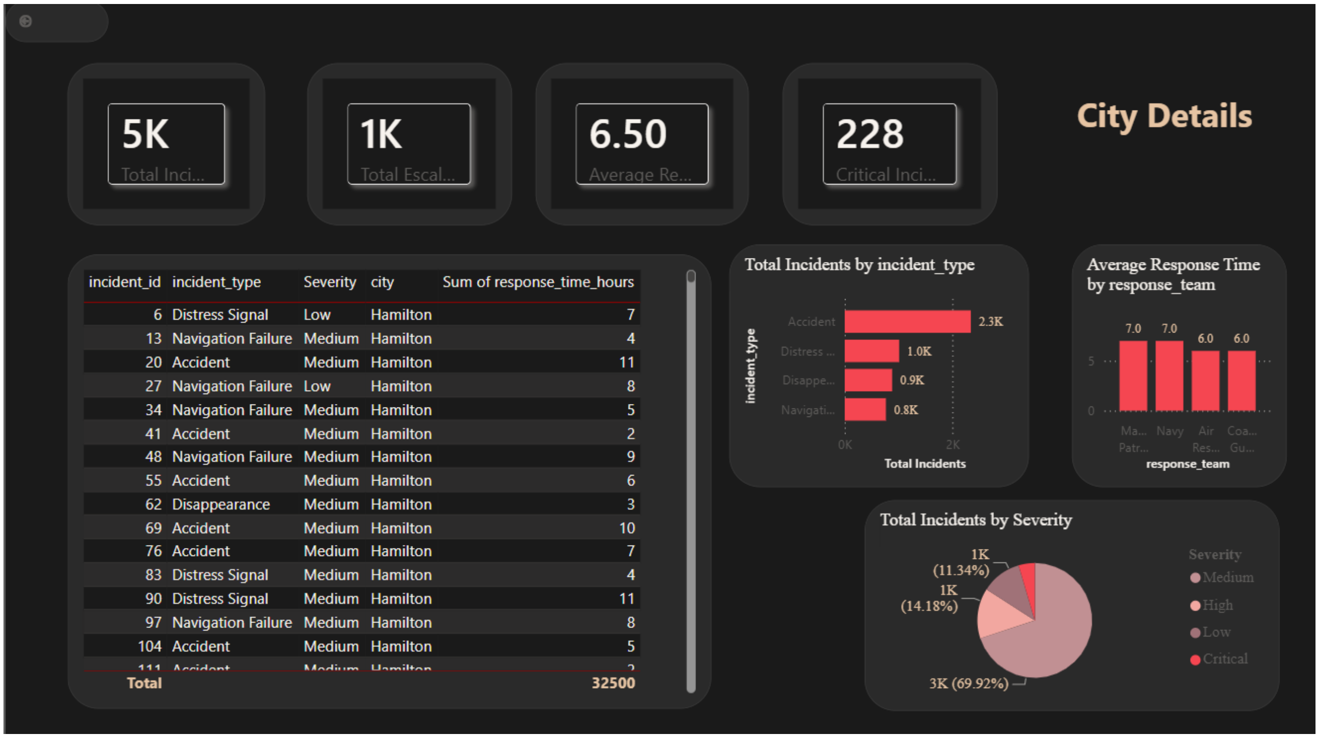
Milestone 2 enhanced the existing dashboard by adding threat classification and geospatial intelligence. The dashboard combined incident, location, and response information to analyze severity, geographical distribution, emergency response performance, and drill-through information.

Major Features

- Threat severity classification
- City-wise incident analysis
- Geographical visualization
- Response performance analysis
- Drill-through reports
- KPI-based monitoring

Key Insights

- **Medium Severity is dominant:** Medium Severity incidents form the largest group in the dataset and are the most frequent.
- **Accident is the most common type:** Accident has the highest recorded incident count compared with Distress Signal, Disappearance, and Navigation Failure.
- **Incidents are concentrated in monitored cities:** The geographical analysis helps identify cities with higher incident counts and compare response performance between locations.



Milestone 3: Temporal & Incident Response Analytics

Objective

Milestone 3 extended the existing solution with Temporal Incident Analytics and Incident Response Analytics. The objective was to understand when incidents occur and evaluate the efficiency of emergency response operations.

Temporal Analysis

- Monthly incident trends
- Seasonal changes
- Daily trends
- Weekly trends
- Activity spikes
- Emerging threats

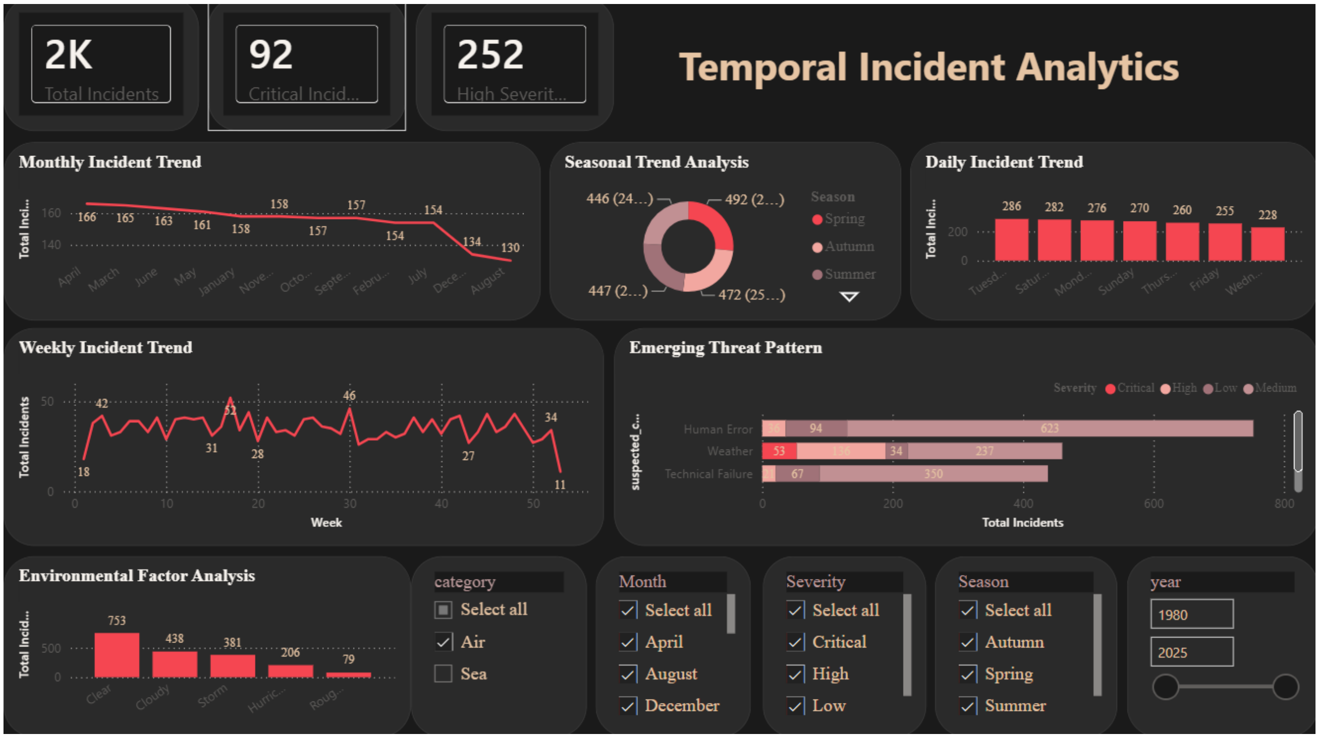
Major emerging threat categories included: Human Error, Weather, and Technical Failure.

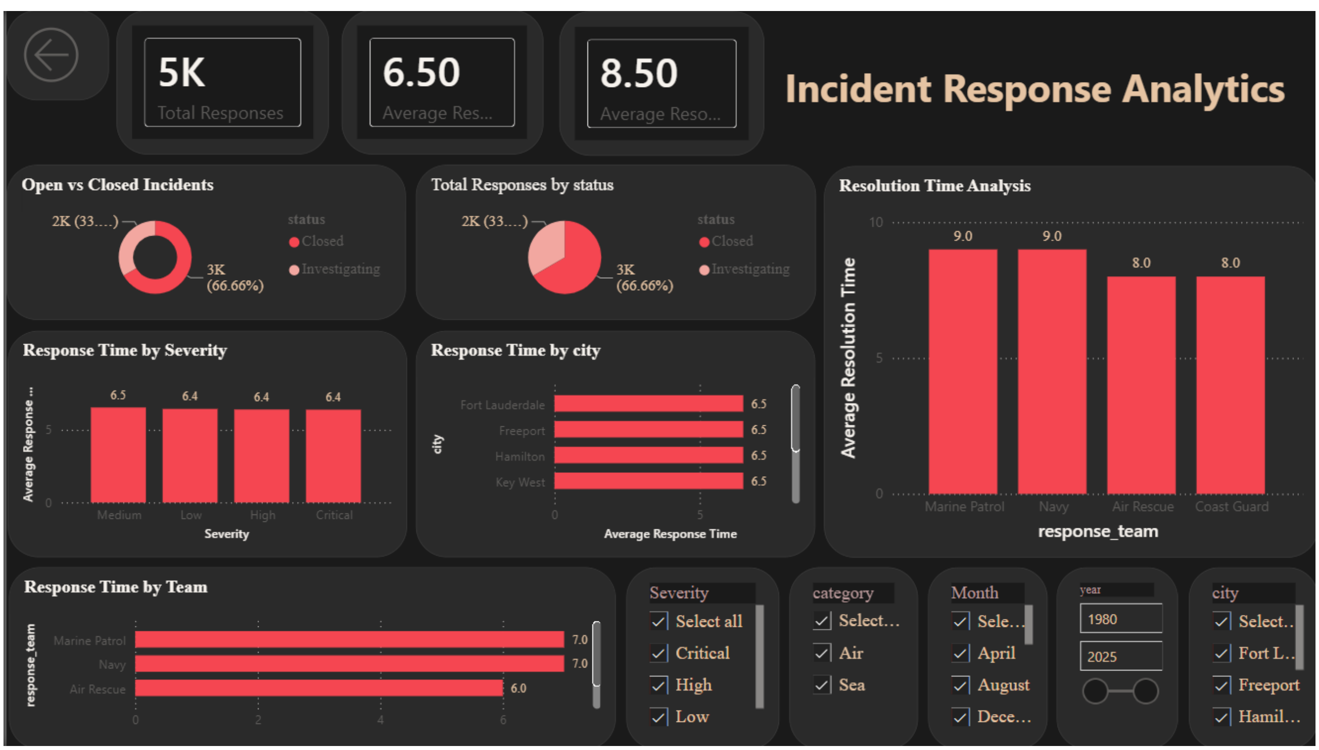
Response Analysis

- Average response time
- Resolution time
- Team performance
- City performance
- Severity-based performance

Key Insights

Incident frequency changes across different time periods. Seasonal and activity-spike analysis can reveal recurring patterns, while response analysis helps identify differences in operational performance.





Milestone 4: Security Risk & Predictive Intelligence

1. Objective

Milestone 4 moves the project beyond historical analysis toward current security risk assessment and predictive intelligence.

The objective is to help stakeholders measure threat exposure, identify operational vulnerabilities and resource gaps, compare risks across locations and operational units, predict emerging risks, and support executive-level decision-making.

2. Security Risk Assessment

The Security Risk Assessment component focuses on:

- Threat Exposure Level
- Operational Vulnerability
- Risk Indicators by Location
- Risk Indicators by Operational Unit
- Resource Allocation Gap
- Security Weakness Analysis
- Security Risk Scorecard

This provides a structured view of security conditions and helps identify areas requiring attention and better resource planning.

3. Predictive Intelligence

- Predictive Threat Risk Model
- Incident Risk / Probability Prediction
- Forecasting of Emerging Risk
- Forecasting of Incident Trends
- Identification of Potential High-Risk Zones
- Model Performance Evaluation

The purpose is to use the available analytical information to support proactive planning rather than relying only on historical reporting.

4. Executive Intelligence

The executive dashboard focuses on high-level decision-support information:

- Overall Risk Score
- High-Risk Locations
- Critical Risk Indicators
- Predicted Risk
- Key Operational Risk Indicators
- Strategic Decision-Support Metrics

5. Dashboard Interactivity

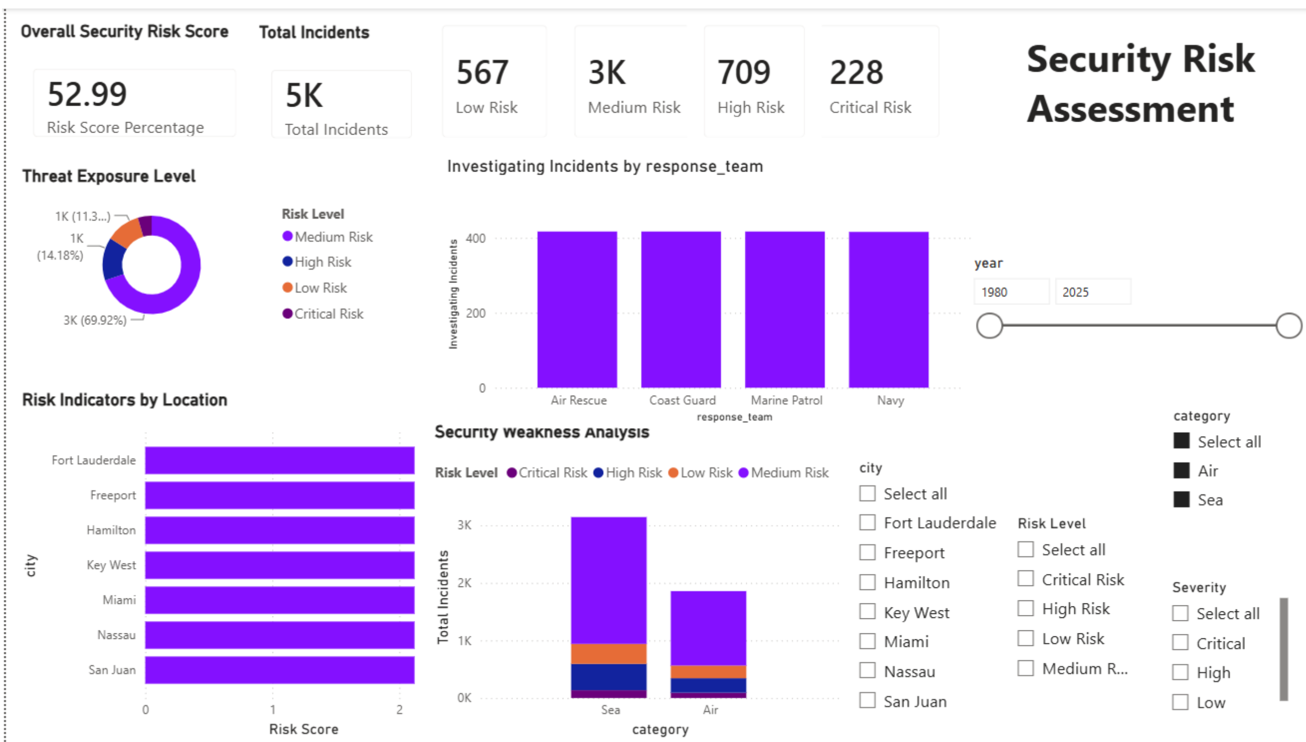
The final solution provides interactive filtering using:

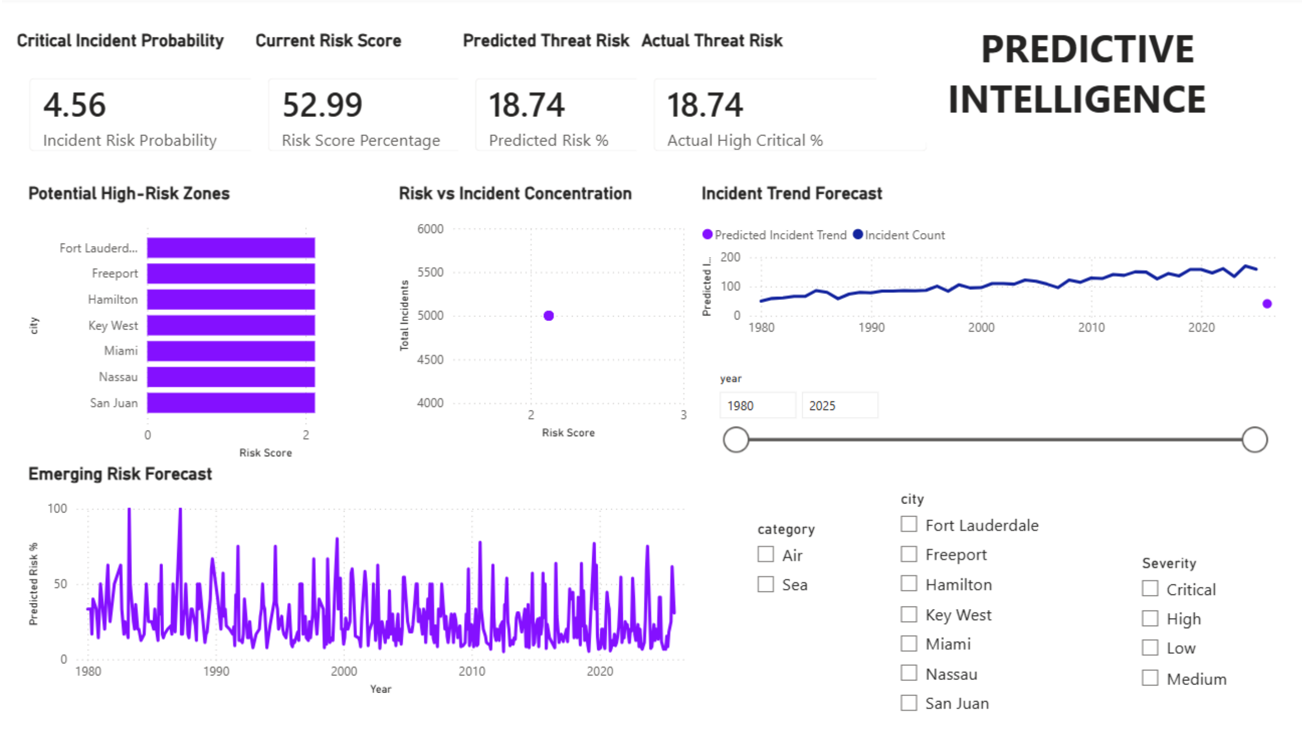
- Year
- Location/City
- Severity
- Category
- Operational Unit
- Risk Level

These filters allow stakeholders to explore the dashboard according to their specific analytical requirements.

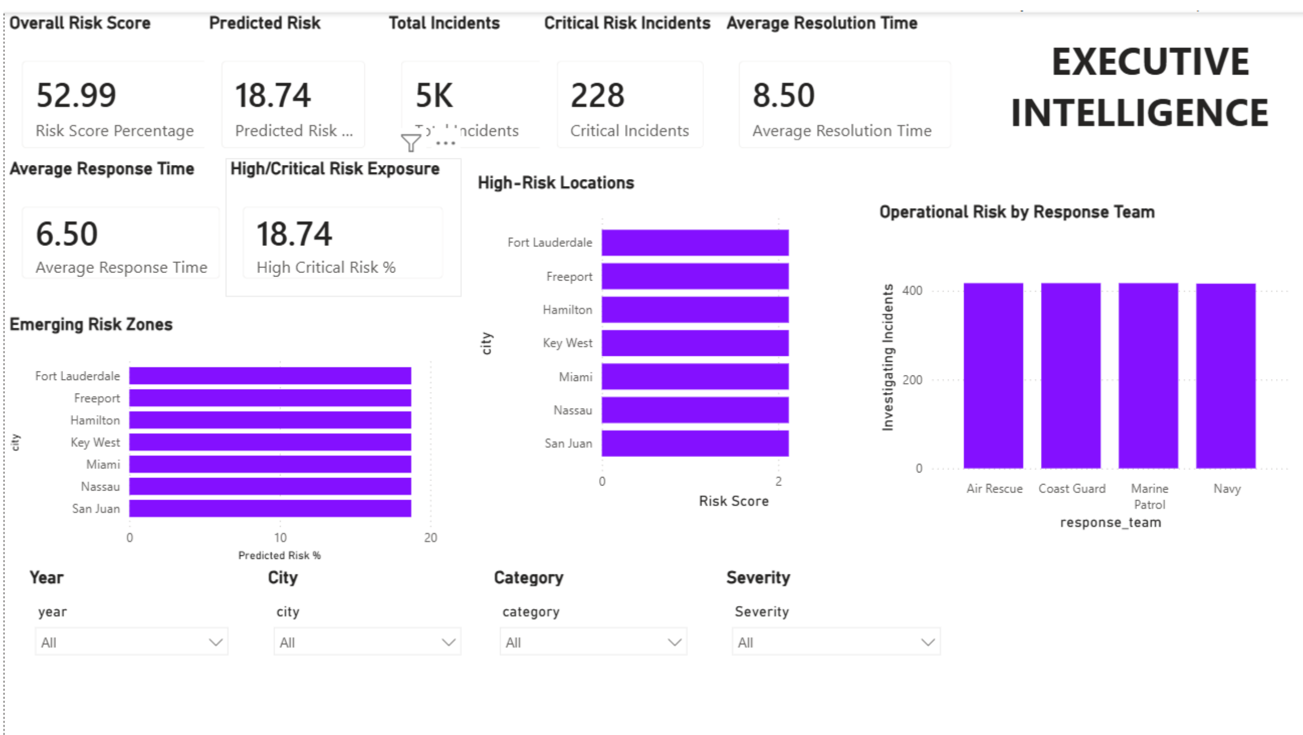
6. Overall Outcome

Milestone 4 completes the transition from historical incident analysis to risk-focused and predictive decision support.





Critical Incident Probability
4.56
Incident Risk Probability
Current Risk Score
52.99
Risk Score Percentage
Predicted Threat Risk
18.74
Predicted Risk %
Actual Threat Risk
18.74
Actual High Critical %
PREDICTIVE INTELLIGENCE
Potential High-Risk Zones
Fort Lauderd...
Freeport
Hamilton
Key West
Miami
Nassau
San Juan
city
0
2
Risk Score
Risk vs Incident Concentration
6000
5500
5000
4500
4000
Total Incidents
2
3
Risk Score
Incident Trend Forecast
Predicted Incident Trend
Incident Count
200
100
0
1980
1990
2000
2010
2020
year
1980
2025
Emerging Risk Forecast
100
50
0
Predicted Risk %
1980
1990
2000
2010
2020
Year
category
Air
Sea
city
Fort Lauderdale
Freeport
Hamilton
Key West
Miami
Nassau
San Juan
Severity
Critical
High
Low
Medium



Overall Risk Score
52.99
Risk Score Percentage
Predicted Risk
18.74
Predicted Risk ...
Total Incidents
5K
Critical Risk Incidents
228
Critical Incidents
Average Resolution Time
8.50
Average Resolution Time
EXECUTIVE INTELLIGENCE
Average Response Time
6.50
Average Response Time
High/Critical Risk Exposure
18.74
High Critical Risk %
High-Risk Locations
Fort Lauderdale
Freeport
Hamilton
Key West
Miami
Nassau
San Juan
city
0
2
Risk Score
Operational Risk by Response Team
Investigating Incidents
400
200
0
Air Rescue
Coast Guard
Marine Patrol
Navy
response_team
Emerging Risk Zones
Fort Lauderdale
Freeport
Hamilton
Key West
Miami
Nassau
San Juan
city
0
10
20
Predicted Risk %
Year
year
All
City
city
All
Category
category
All
Severity
Severity
All

1. Major Project Insights

Across the four milestones, the project provides the following major insights:

Incident Patterns

Incident data can be analyzed by type, category, severity, location, and outcome to understand overall threat patterns.

Severity

Medium severity incidents are the most frequent, while the dashboard also identifies High and Critical incidents requiring greater attention.

Incident Type

Accident is the most common incident type in the analyzed dataset.

Location

Incident activity is concentrated across monitored cities, allowing high-incident locations to be identified and compared.

Time and Response

Incident frequency varies across different time periods, while response performance differs across teams, cities, and severity levels.

2. Strategic Recommendations

- **Improve Resource Allocation —** Allocate additional resources during periods and locations identified as having higher incident activity.
- **Improve Response Performance —** Review response and resolution performance in locations or operational teams with higher response times.
- **Monitor Emerging Risks —** Use interactive and predictive analytics to continuously monitor emerging threat patterns and support operational planning.

3. Final Conclusion

The project successfully developed an interactive Power BI solution that progresses through four stages of analytics: incident analysis, geospatial intelligence, temporal and response analytics, and predictive risk intelligence.

The dashboards combine data modeling, DAX calculations, KPI cards, charts, maps, slicers, and analytical views to transform incident data into meaningful insights.

The final solution supports stakeholders in understanding existing threats, identifying risk areas, evaluating operational performance, and using predictive intelligence for better planning and decision-making.

4. Project Deliverables

Deliverable	File Name
Power BI Dashboard	Threat-Classification-Predictive.pbix
Project Documentation	Project-Documentation.pdf
Final Presentation	Final-Presentation.pptx